

GREENHOUSE EFFECT COMPARISON LAB



Lesson

This lesson introduces the greenhouse effect and explains how certain gases in Earth's atmosphere help trap heat. Students learn that sunlight can enter the atmosphere easily, but some of the heat that Earth gives off can be held in by gases such as carbon dioxide. This trapped heat helps keep Earth warm enough for life, but too much carbon dioxide can cause the planet to warm more than normal. Students connect this idea to climate change by discussing how burning fossil fuels adds extra carbon dioxide to the air.

Experiment

In the Greenhouse Effect Comparison experiment, students test whether air with extra carbon dioxide warms differently than normal air. Students set up two clear bottles with thermometers and seal them so heat cannot easily escape. One bottle is filled with normal air, and the other is filled with carbon dioxide made by mixing baking soda and vinegar. Both bottles are placed under a bright lamp or in direct sunlight, and students record the temperature in each bottle every one to two minutes for about fifteen to twenty minutes. After collecting data, students compare the temperature changes and use their results as evidence to explain how carbon dioxide can increase heat trapping.

